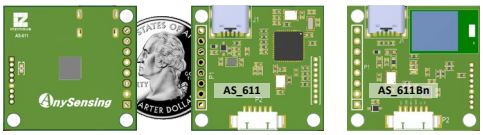
AS_611B / AS_611

Distance, Human and Animal's Vital-sign detection Radar
with High performance, Small size, Low-power
GPIO, UART, BLE, Indicate LED

User Manual

1 Features

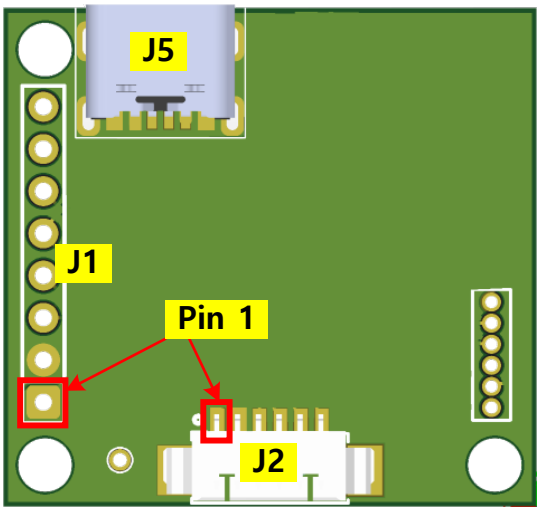


- 60GHz pulsed radar
- One board solution
- Include antenna and DSP module
- Wide input voltage available: 3.3~7.0V
- Small size and easy to use
- GPIO, UART, BLE(AS_611B only) interface
- Depending on the function provided, it is divided into a distance detection model and a vital-sign detection model
- Type of vital-sign provided: Human - BR(Breath rate), BR waveform, HR(Heart Rate), HR waveform, HRV(RMSSD, SDNN)
- Detection of more than 85% accurate vital-signs in steady state. At a steady state, the respiratory rate is more than 90% accurate
- 3 types of power source input possible: USB-C, 2.54mm pin-header, connector

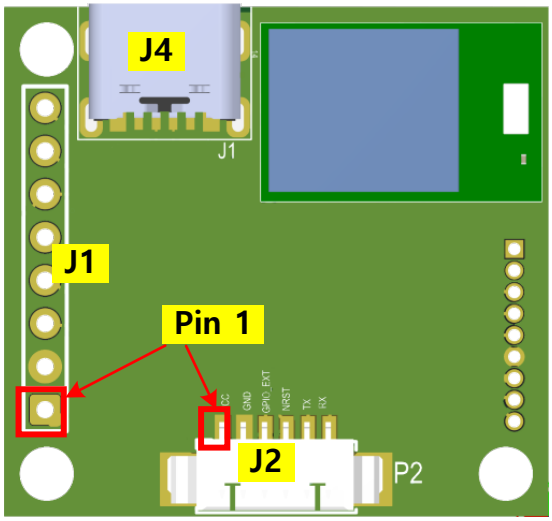
2 Specification

Content		Definition
Model name		AS_611Bn : BLE-enabled model (basic model) AS_611 : MCU Based model
Hardware size		32.0 x 30.0 x 3.35 mm (No connector) 32.0 x 30.0 x 5.1 mm (Applying connector)
Weight		3.60g
Transmit frequency		60GHz
Antenna		1 TX (EIRP : 10dBm), 1 RX / Antenna gain : 2dBi
RF pattern (-3dB width)		Horizontal 65° / Vertical 55°
Supply voltage / current		3.3 ~ 7 VDC / 0.3W
Operating / Storage temperature		-40 ~ 85 ℃ / -55 ~ 125 ℃
Interface		GPIO, UART(TTL level), BLE(just AS_611Bn)
Function		Distance, Vital-sign(BR, HR, HRV), Indicate LED
Distance	Range	Distance : 0.1 ~ 20 m (set by zmin / zmax, default 0.25 ~ 3.0 m) Humans guarantee detection up to 4 m. Vital-sign : 0.15 ~ 2.5 m (set by zmin / zmax, default 0.25 ~ 1.5 m)
	Resolution	Vital sign : under 1.0 m more than 85% over 1.0 m more than 80%

3 Pin description



AS_611

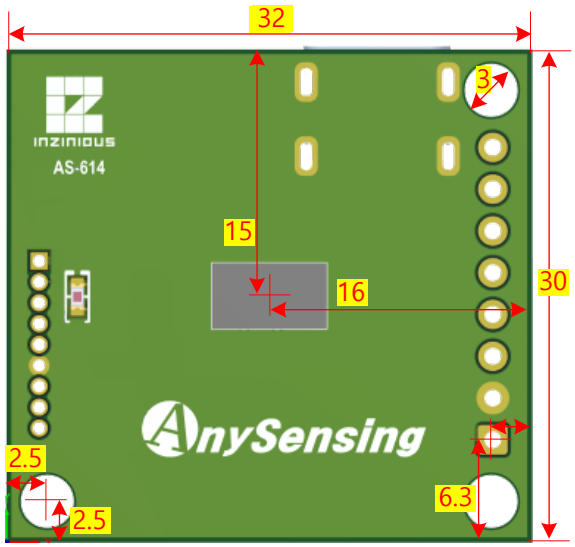


AS_611Bn

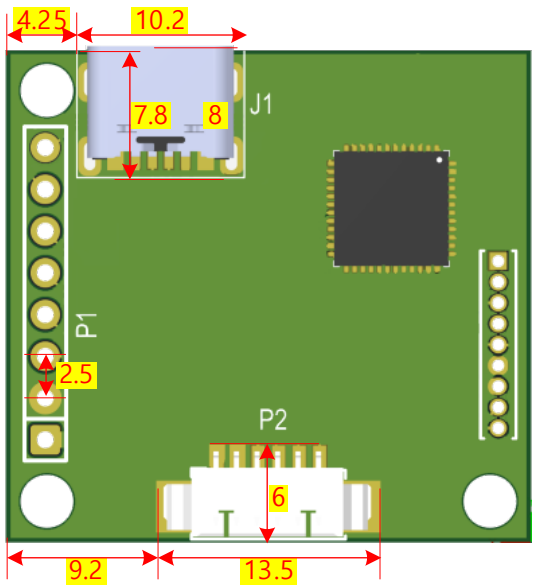
PIN	Name	J1	J2
1	VCC	Power + (3.3 ~ 7V)	Power + (3.3 ~ 7V)
2	GND	Power -	Power -
3	GPIO	In/out	In/out
4	NRST	N.C or High : Normal operation Low : Reset	N.C or High : Normal operation Low : Reset
5	UART TX 1	UART 1 data transmit	UART data transmit
6	UART RX 1	UART 1 data receive	UART data receive
7	UART TX 2	UART 2 data transmit	Not used
8	UART RX 2	UART 2 data receive	Not used

J1 : 2.54mm Pin header
J2 : 12505WR-06, The connecting connector is 12505HS-06.
J4, J5 : USB-C type connector only supplies power

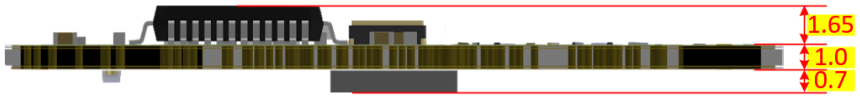
4 Dimension



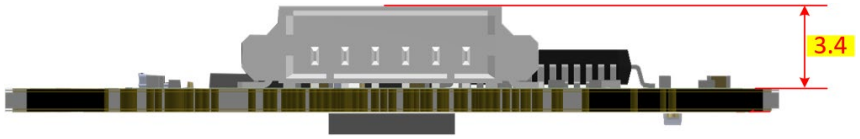
Top



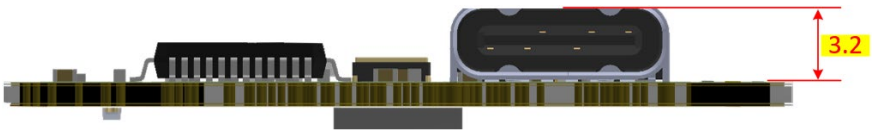
Bottom



No connector



Install connector (12505WR-06)



Install USB-C connector

5 Function

Function Model	Mode	Function
Distance detection	**D**istance	Mode that detects the distance of an object Can detect motionless objects Detects multiple objects (up to 10)
Vital-sign detection	**V**ital-sign	Mode that detects vital-signs from a person Output breath rate, heart rate, heart/breath waveform and HRV Detection up to 2.5 m

6 Distance detection model

- The distance detection model measures and outputs the distance and the reflected signal strength of each detected object. The measurement range is freely configurable between the minimum (zmin) and maximum (zmax) distance, in centimeters.
- The maximum measurement distance can be set up to 20 m (zmax, default 3.0 m) and the minimum down to 0.1 m (zmin, default 0.25 m). The Acconeer A121 distance detector automatically optimizes the profile, step length and PRF for the configured range. Multiple objects can be detected at once.
- The supported features are as follows:
  - Configuration through UART / BLE commands
  - Measurement range setting: 0.1 m ~ 20 m (zmin / zmax, in cm)
  - Detects both stationary and moving objects
  - Multiple object detection, output sorted by distance or by strength (sort)
  - Detection sensitivity adjustment via CFAR threshold (sena)
  - Performance mode selection: balanced / fast / accurate (pmode)
  - Output timing: real-time (15 Hz) or once per second (outtime)
  - Fixed ASCII frame output format
  - Settings stored in non-volatile memory (retained after power-off)
  - Software and hardware reset (reset / freset)

- Measurement range and default settings

Item	Default	Setting range
Minimum distance (zmin)	25 cm	10 ~ (zmax - 25) cm
Maximum distance (zmax)	300 cm (3.0 m)	50 ~ 2000 cm (20 m)
Sensitivity (sena)	50 %	0 ~ 100 %
Performance (pmode)	0 (BALANCED)	0:BALANCED, 1:FAST, 2:ACCURATE
Output sort (sort)	0 (by distance)	0:distance, 1:strength

- **Notes on detection performance and options**
  - In an open environment with few surrounding reflectors the detection performance is high. Many nearby obstacles degrade detection.
  - The 'sena' option sets the CFAR detection sensitivity (0 ~ 100 %). A higher value detects weaker or smaller objects, but increases the chance of false detection from surrounding clutter.
  - The 'pmode' option selects the performance profile - BALANCED (0, default), FAST (1, faster update), ACCURATE (2, better range/precision).
  - The 'sort' option selects the output order of multiple objects: by distance (closest first) or by reflected strength (strongest first).
  - The 'outtime' option selects the output timing: real-time (15 Hz) or once per second.
  - The 'zmin' and 'zmax' options set the minimum and maximum measurement distance in centimeters. Narrowing the range to the area of interest improves stability and rejects unwanted reflections.
  - All settings are stored in non-volatile memory and restored automatically after a power cycle.
  - Depending on the usage environment the options should be set appropriately to achieve the best detection performance.

- **Recommended configuration sequence**

Sequence	Command	Description
1	mode 1	Select DISTANCE mode (1: DISTANCE, 2: VITAL). Default: DISTANCE.
2	zmax n	Set the maximum measurement distance in cm (50 ~ 2000). Default 300.
3	zmin n	Set the minimum measurement distance in cm (10 ~ zmax-25). Default 25.
4	pmode n	Select the performance mode. 0:BALANCED, 1:FAST, 2:ACCURATE. Default 0.
5	sena n	Set the CFAR sensitivity (0 ~ 100 %). Higher detects weaker objects. Default 50.
6	sort n	Select the output order. 0: by distance, 1: by strength. Default 0.
7	outtime n	Select the output timing. 0: real-time (15 Hz), 1: once per second. Default 1.

- Data Output Format : ASCII
  - Distance output format

Output format		#ABBBBCCC! (10 byte)
Item	**Byte**	**Content**
#	1	Character that starts the distance data output
A	1	Object number (1: first object, 2: second object, ...)
BBBB	4	Distance from the radar to the object, unit in cm
CCC	3	Reflected signal strength of the detected object
!	1	Character indicating the end of data

Example: object 1 at 250 cm with strength 45 is output as #10250045!
When no object is detected, '#0!' is output every 15 seconds.

- On boot or reset the system outputs '#AS_611B_4.0!'.

7 Vital sign detection model

- Detecting steady-state human vital-signs (breath, heart rate, HRV)
  - Only human detection is supported. The module measures the vital-signs of one person located in front of the radar.
  - Over 80 ~ 90% accuracy in steady state
  - The detection distance is configurable from 0.15 m to 2.5 m via zmin / zmax (in cm); the default range is 0.25 m ~ 1.5 m. Detection angle is approximately 70 deg.
  - The measurement range is set by the minimum (zmin) and maximum (zmax) distance.
  - Detectable motion: basically, the level of mouse manipulation or simple conversation
  - Vital-sign detection is continuously provided even during simple conversations or small movements, but the accuracy is lower than in a stable state.
  - Only one person in front of the radar can be detected
  - Vital-signs can be detected from the front, side and back of a person, and measuring from the front ensures the most accurate results
- The output content and rate are selected with the 'wave' command; by default (wave 0) the module is in simple output mode.
- Simple output mode (wave 0, default) outputs the heart rate, respiration rate and distance once per second. Waveform output mode (wave 1) additionally outputs the heart-beat pulse and respiration waveform, 10 times per second. The 'exten' command adds HRV (RMSSD, SDNN) to either mode.
- Does not guarantee normal detection if there are other people near the detecting person
- **In the following cases, detection is not possible or accuracy is low**.
  - When there is another object between the radar and the person detecting it, or when there is an object such as metal or a moving object nearby.
  - In particular, when the maximum measurement distance (zmax) is set to 1.5 m or more, vital-signs may be missed if highly reflective objects are near the radar or the person. Reducing zmax

to cover only the area of interest improves this.

- When a person is leaning on an object such as a sofa that has high tension
- Intentionally holding your breath When the radar is not directly facing the chest, but is next to the person and the arm blocks the radar signal
- When there is a moving object near a person
- When more than one person is together

- **Detection of the heart rate signal captures the movement of a very small chest area, so detection accuracy can be ensured when the subject is in a stable state. Reliable detection results are guaranteed when the detection starts and the subject remains stable for more than 20 seconds.**

- Data Output Format : ASCII
  - ■ Simple output mode (wave 0, default)

Output format		%ABBBCCDDD! (10 byte)
Item	**Byte**	**Content**
%	1	Start character of simple vital-sign data (wave off)
A	1	Vital-sign status 0: No target 1: Target, moving 2: Target, calculating 3: Target, vital-sign
BBB	3	Heart rate per minute (bpm)
CC	2	Respiration rate per minute (bpm)
DDD	3	Distance to the target, unit in cm
!	1	Character indicating the end of data

With extension data enabled (exten 1) the simple format becomes %ABBBCCDDEEFFF! - DD: HRV RMSSD (ms), EE: HRV SDNN (ms), FFF: distance (cm).

  - ■ Waveform output mode (wave 1)

Output format		@ABBBCCDEEEFFF! (14 byte)
Item	**Byte**	**Content**
@	1	Start character of waveform vital-sign data (wave on)
A	1	Vital-sign status 0: No target 1: Target, moving 2: Target, calculating 3: Target, vital-sign
BBB	3	Heart rate per minute (bpm)
CC	2	Respiration rate per minute (bpm)
D	1	Heart-beat pulse (1 at each beat, otherwise 0)
EEE	3	Respiration waveform data (0 ~ 999)
FFF	3	Distance to the target, unit in cm
!	1	Character indicating the end of data

With extension data enabled (exten 1) the waveform format becomes @ABBBCCDDEEFGGGHHH! - DD: HRV RMSSD (ms), EE: HRV SDNN (ms), F: heart pulse, GGG: respiration waveform, HHH: distance (cm).

Waveform mode outputs 10 times per second; simple mode outputs once per second.

- When no target is detected, '%0!' (simple mode) or '@0!' (waveform mode) is output every 15 seconds. On boot or reset the system outputs '#AS_611B_4.0!'.

8 Protocol

8.1 Communication Method

- AS_611Bn supports 3.3V level UART communication, AS_611B supports 3.3V level UART and BLE-based UART communications
- Utilizes ASCII character codes for sending and receiving data and supports Command Line Input (CLI)

- The LED on the radar module flashes when data is output.
- UART Settings

Item	Setting value
Baud rate	460,800 bps (default) Configurable: 115200 / 230400 / 460800 / 921600 (baud)
Data	8 bit
Parity bit	None
Stop bits	1 bit
Flow control	None

- All commands must be in lowercase
- When a setting that triggers reconfiguration is changed, the data output briefly stops while the module re-initializes. After a 'baud' change the module reboots, so the BLE connection must be re-established.

8.2 Distance detection model

Command	Example	Default value / Setting range	Explanation
mode	mode 1	1 / 1, 2	Select the operation mode. 1: DISTANCE, 2: VITAL. Returns =DISTANCE! or =VITAL!
start	start	-	Start data output.
stop	stop	-	Stop data output.
zmin	zmin 25	25 / 10 ~ (zmax-25) cm	Minimum measurement distance, in cm.
zmax	zmax 300	300 / 50 ~ 2000 cm	Maximum measurement distance, in cm (up to 20 m).
pmode	pmode 0	0 / 0, 1, 2	Performance mode. 0: BALANCED, 1: FAST, 2: ACCURATE.
sena	sena 50	50 / 0 ~ 100	CFAR detection sensitivity (%). A higher value detects weaker / smaller objects but increases the chance of false detection.
sort	sort 0	0 / 0, 1	Output order of detected objects. 0: by distance (closest first), 1: by strength (strongest first).
outtime	outtime 1	1 / 0, 1	Output timing. 0: real-time (15 Hz), 1: once per second.
baud	baud 460800	460800 / 115200, 230400, 460800, 921600	UART baud rate. Saved to flash and applied after an automatic reboot; the host must reconnect at the same rate.
reset	reset	-	Software reset. Re-initializes the current mode and returns to output.
freset	freset	-	Hardware (cold) reset of the module.
test	test 0	0 / 0, 1	Test / debug output mode. 0: Normal, 1: Wave data.

8.3 Vital sign detection model

Command	Example	Default value / Setting range	Explanation
mode	mode 2	1 / 1, 2	Select the operation mode. 1: DISTANCE, 2: VITAL. Returns =DISTANCE! or =VITAL!
start	start	-	Start data output.
stop	stop	-	Stop data output.
zmin	zmin 25	25 / 15 ~ (zmax-25) cm	Minimum detection distance, in cm.
zmax	zmax 150	150 / 40 ~ 250 cm	Maximum detection distance, in cm (up to 2.5 m).
wave	wave 1	0 / 0, 1	Output mode. 0: simple (BR / HR / distance once per second), 1: waveform (adds heart-beat pulse and respiration waveform, 10 Hz). VITAL mode only.
exten	exten 1	0 / 0, 1	Extension data. 0: Normal, 1: Extension (adds HRV RMSSD and SDNN to the output).
baud	baud 460800	460800 / 115200, 230400, 460800, 921600	UART baud rate. Saved to flash and applied after an automatic reboot.
reset	reset	-	Software reset. Re-initializes the current mode and returns to output.
freset	freset	-	Hardware (cold) reset of the module.
test	test 0	0 / 0, 1	Test / debug output mode. 0: Normal, 1: Wave data.

9 Bluetooth Information

AS_611Bn supports Bluetooth 4.0 and later. To check the transmitted and received data over Bluetooth, install the [nRF Toolbox] app on Android or iOS and select the UART (Universal Asynchronous Receiver/Transmitter) item under Utils services. When AS_611Bn is powered, it advertises as 'AS_611B' and can be found with the scanner.

Note: If you change the device's settings, the device will reboot again. Therefore, Bluetooth needs to be reconnected as well.

9.1 Bluetooth settings for app/web development

The AS_611B is compliant with the BLE standard GATT and uses the Nordic UART Service.

Service	Value
Service	Nordic UART Service
Vendor-specific UUID	6E400001-B5A3-F393-E0A9-E50E24DCCA9E
TX UUID	6E400003-B5A3-F393-E0A9-E50E24DCCA9E
RX UUID	6E400002-B5A3-F393-E0A9-E50E24DCCA9E

10 Order Information

Order model : AS_611(Bn)_D/V_P/C/U
AS_611(Bn) _ Function model _ Interface select

- AS_611 : Based model
- AS_611Bn : BLE support model
- D/V : D : Distance detection function model
  : V : Vital sign detection function model
- P/C/U : P : Installing 2.54mm pin header
  : C : Installing connector
  : U : Installing USB-C connector

Ex) AS_611_D : Based model of distance sensing model without connectors or pin header
AS_611Bn_V_U: Vital sign detection BLE support model with USB-C connector
AS_611_V_CU : Based model of vital sign detection with USB-C connector and connector

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